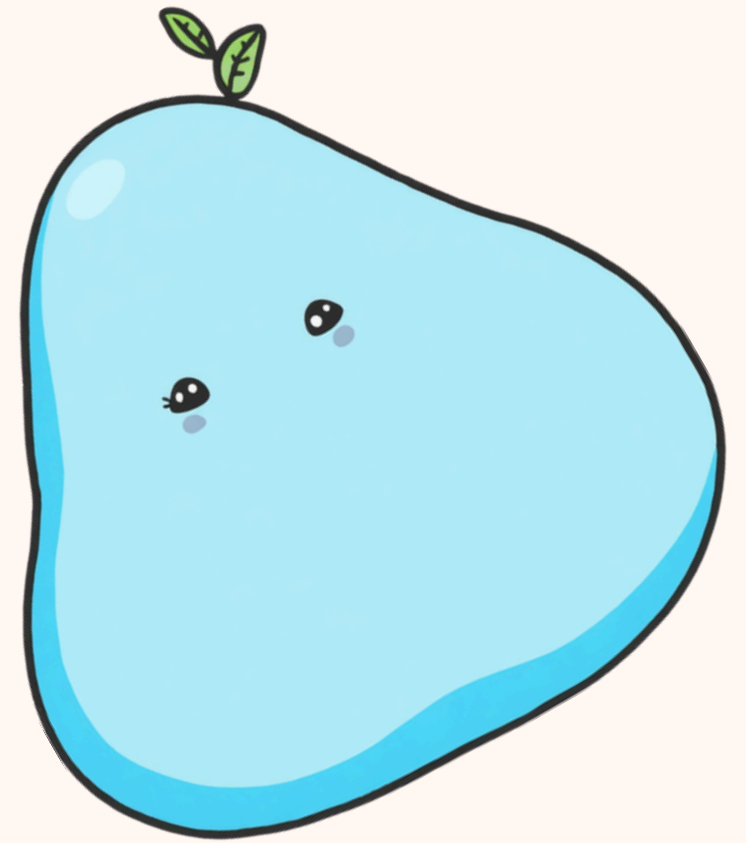
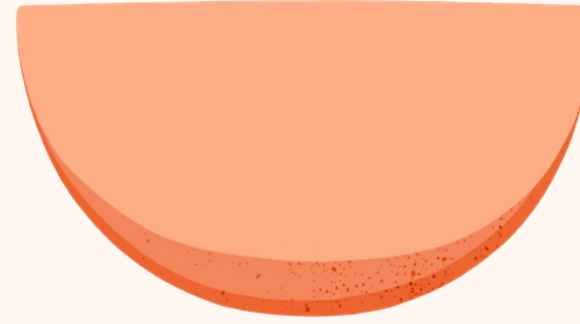




DATA STRUCTURES STACKS & QUEUES: SORTING LIFE WITH LOGIC



THE MESSY CUPBOARD: WHY ORGANIZE?

Why do we need to organize things? How do we keep our school bags, books, or tiffin boxes neat?

Think about it: What happens when you can't find your homework?



WHAT IS A DATA STRUCTURE?

A Data Structure is a way to organize information so it's easy to find and use. Just like how we arrange things in our school bags or keep our books in order, computers need ways to organize data too!



SIMPLE EXAMPLES

Think about how you organize your lunch boxes, arrange books on a shelf, or stack tiffin dabbas. These are all ways of organizing things!

MEET THE STACK: LAST IN, FIRST OUT

WHAT IS A STACK?

In a Stack, the last thing you put in is the first thing you take out. This is called LIFO — Last In, First Out!

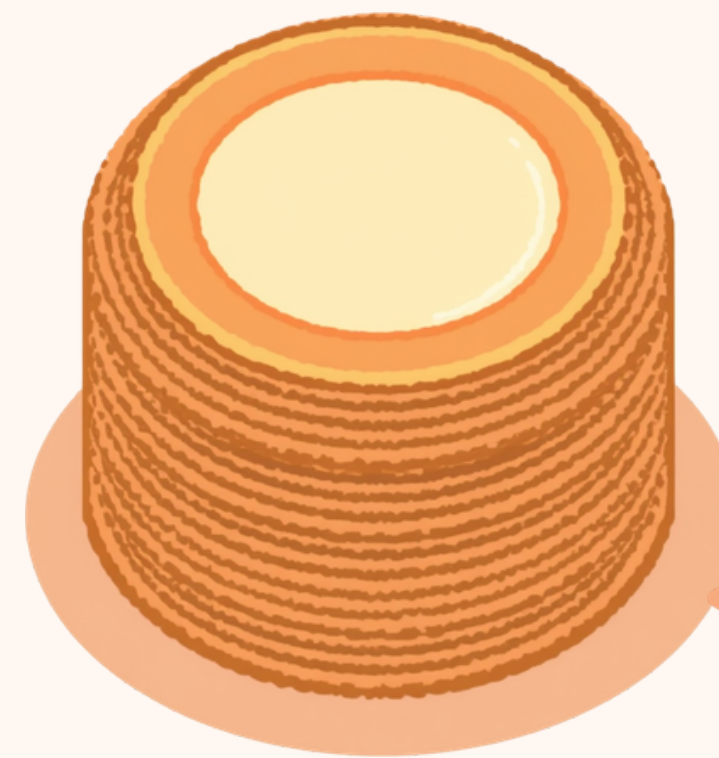
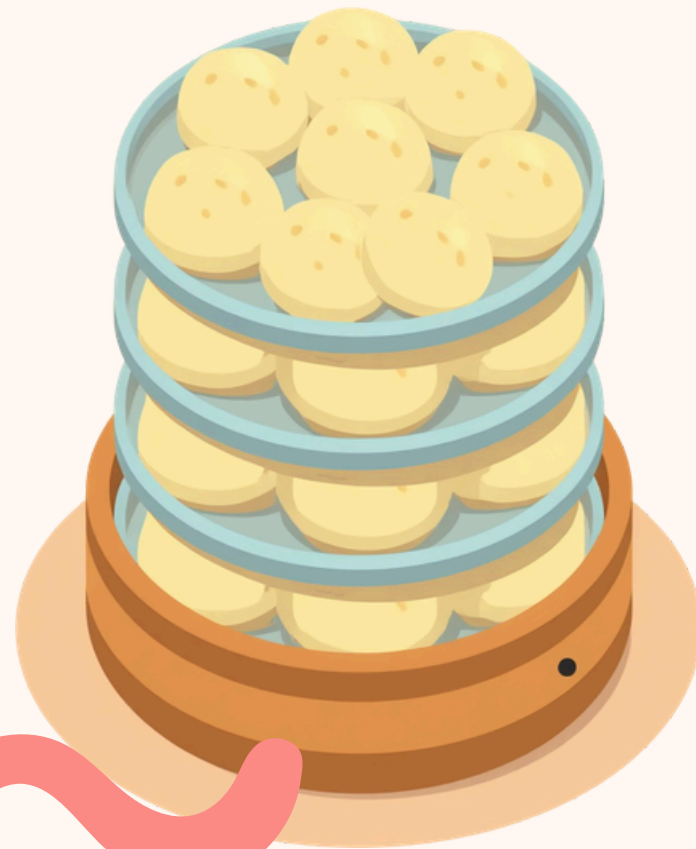
THE LIFO PRINCIPLE

Think of stacking plates at a wedding buffet — the last plate placed on top is always the first one taken!



STACKS AROUND US: INDIAN EXAMPLES

Discover how stacks appear in everyday Indian life! The last item added is always the first one removed — that's the LIFO principle in action.





HOW STACKS WORK: PUSH AND POP

PUSH

Push means adding a new item on TOP of the stack. Like placing a new plate on a pile!



POP

Pop means removing the item from the TOP of the stack. You can only take what's on top first!

REMEMBER!

The last item pushed is always on top. That's why it's called LIFO!



MEET THE QUEUE: FIRST IN, FIRST OUT

WHAT IS A QUEUE?

Think of waiting in line at a railway ticket counter or an auto-rickshaw stand. The person who arrives first gets served first!

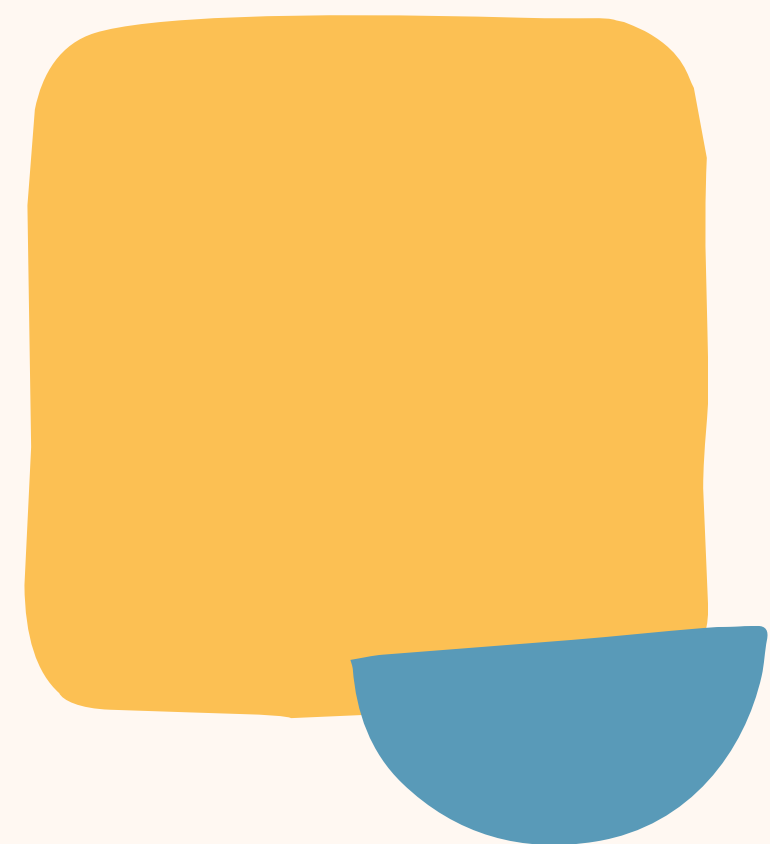
THE FIFO PRINCIPLE

In a Queue, the first person in line is the first to be served. This is called the FIFO principle!



QUEUES AROUND US: INDIAN EXAMPLES

Here are some everyday queues you see in India! In each one, the first person to arrive is the first to be served — that's FIFO!



HOW QUEUES WORK: ENQUEUE AND DEQUEUE

ENQUEUE

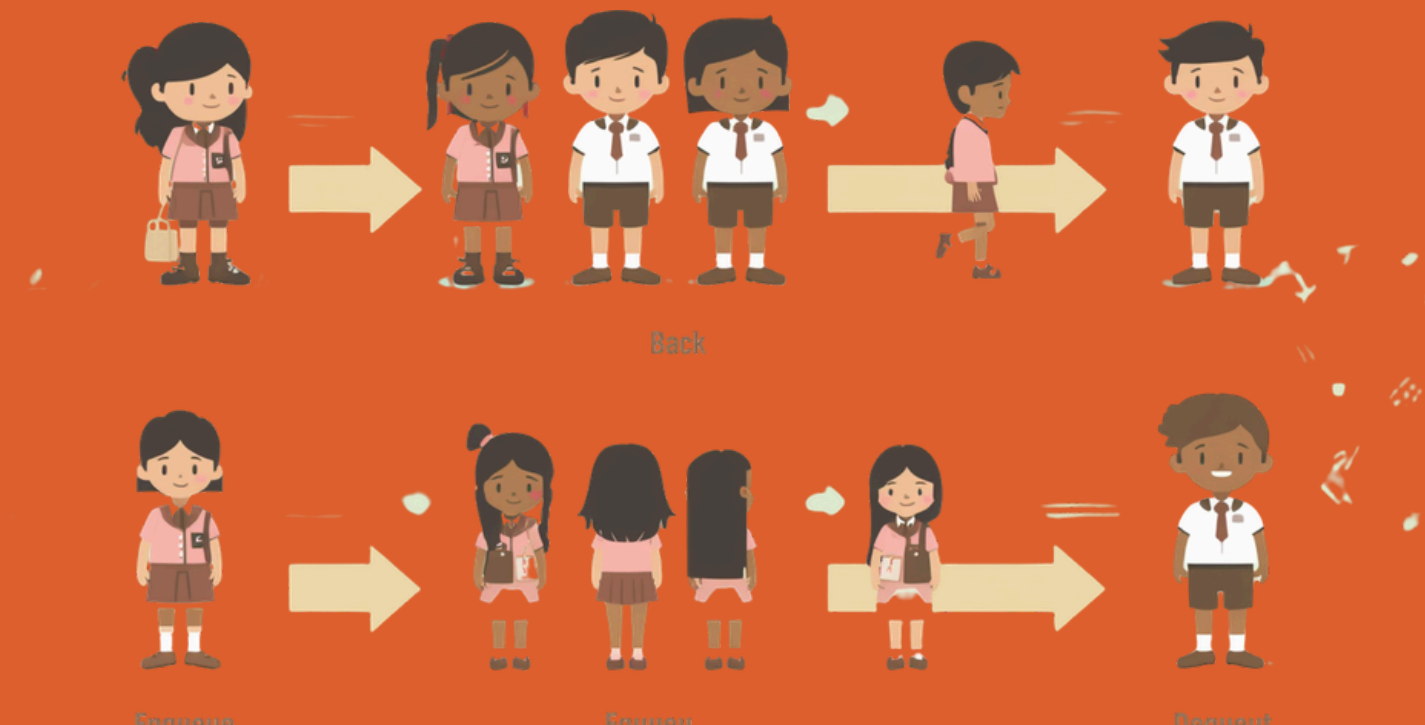
Add a new person or item to the BACK of the line. Like joining the end of a queue!

DEQUEUE

Remove the person or item from the FRONT of the line. The first one in gets served first!

REMEMBER!

FIFO keeps things fair — first come, first served!



INTERACTIVE QUIZ: STACK OR QUEUE?

- PE Class Line
- Queue (FIFO)
- Guess the data structure!

- Pile of Books
- Stack (LIFO)
- Guess the data structure!



INTERACTIVE QUIZ: STACK OR QUEUE? (PART 2)

Scenario 1:

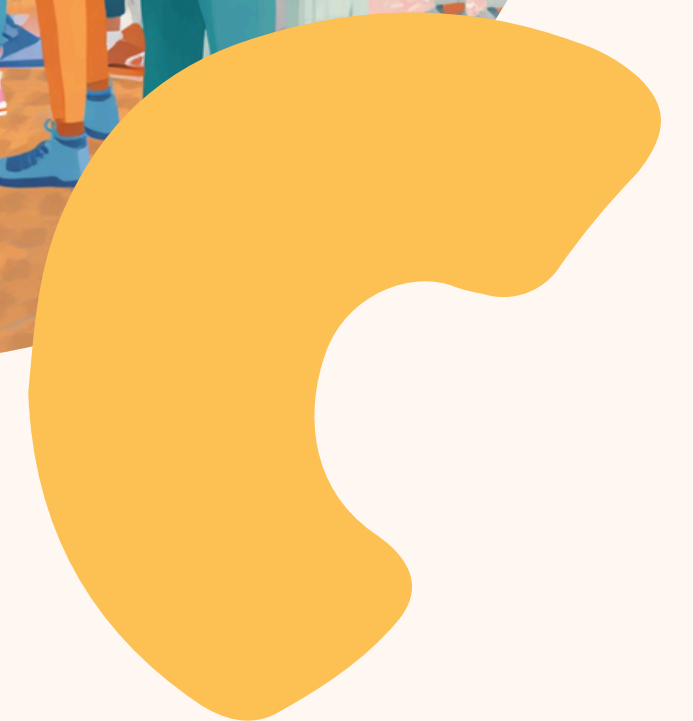
Metro train boarding

Answer: Queue (FIFO)

Scenario 2:

Chapati stack on plate

Answer: Stack (LIFO)



INTERACTIVE QUIZ: STACK OR QUEUE? (PART 3)

Scenario: Pani Puri Stall Line

People waiting at a pani puri stall are served one by one. The person who came first gets their pani puri first!

Is this a Stack or a Queue?

Think about it: Who gets served first — the first person or the last person in line?



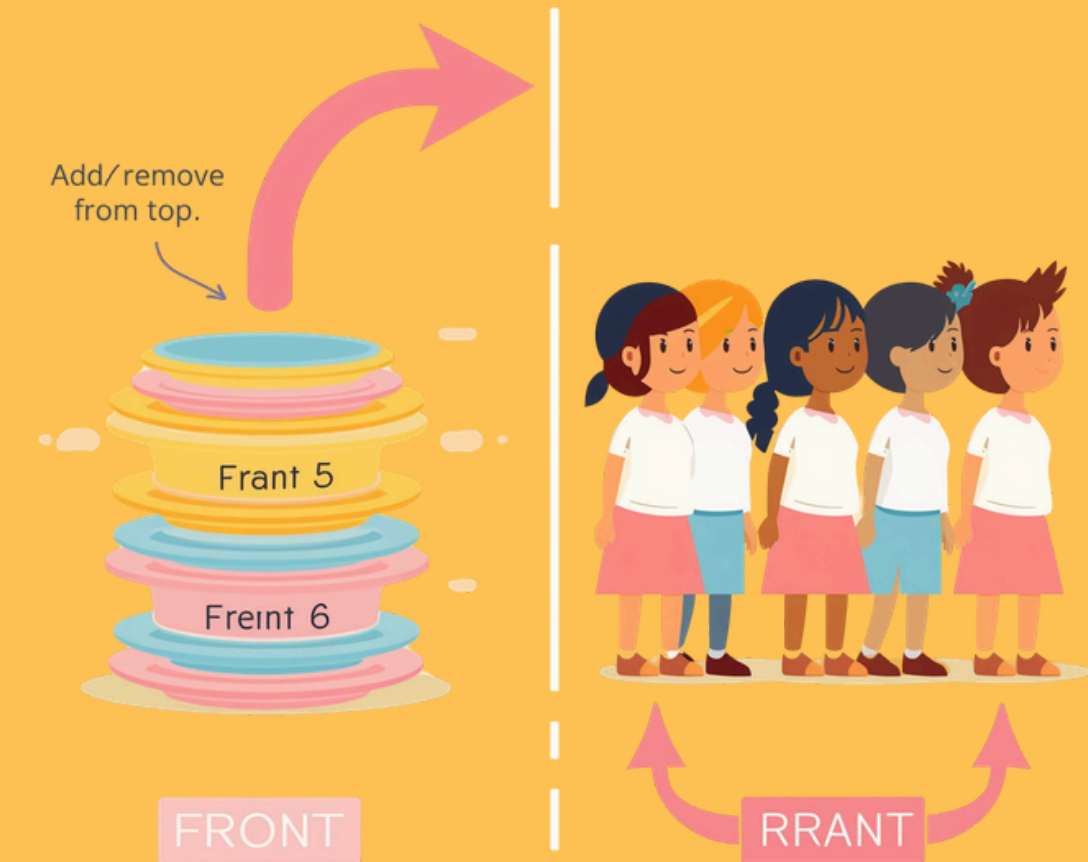
RECAP: LIFO VS FIFO

STACK (LIFO) LAST IN, FIRST OUT

Add and remove from the TOP only. Like a stack of plates — last plate placed is first taken!

KEY DIFFERENCE HOW ITEMS ARE ADDED & REMOVED

Stack = vertical pile. Think idli steamer, bangles on wrist, or notebooks stacked up.



QUEUE (FIFO) FIRST IN, FIRST OUT

Add at REAR, remove from FRONT. Like a queue at the railway counter — first person is served first!



KEY TAKEAWAYS & SUCCESS CRITERIA

STACK SKILLS

Identify the "Top" of a stack and use LIFO (Last In, First Out) correctly.



REAL-WORLD SORTING

Identify the "Front" and "Rear" of a queue using FIFO (First In, First Out).

QUEUE SKILLS

Categorize real-world situations as stacks or queues with 90% accuracy. Remember: These structures help us organize life with logic!





THANK YOU!

Exit Ticket: "Write one sentence: What's the difference between LIFO and FIFO?"

Great job! See you next time for more fun with data structures!